$$G(\omega)=\frac{c\left(1-m\omega-\ln\frac{c}{(\pi m+r)\omega}+(1-m\omega)^{2}\,\omega\left(\frac{c}{(\pi m+r)\omega}\right)^{\frac{1}{1-m\omega}}\right)}{1-m\omega-m\omega\ln\frac{c}{(\pi m+r)\omega}}. \tag{31}$$

$$\begin{aligned}\frac{\partial u_c(\omega)}{\partial\omega}=g\frac{m}{\omega(m\omega-1)^{2}}\left(\frac{c}{(\pi m+r)\omega}\right)^{\frac{1}{m\omega-1}}\left(1-m\omega-m\omega\ln\frac{c}{(\pi m+r)\omega}\right)-cm\\ -\frac{m(\pi m+r)}{(m\omega-1)^{2}}\left(\frac{c}{(\pi m+r)\omega}\right)^{\frac{m\omega}{m\omega-1}}\left(1-m\omega-\ln\frac{c}{(\pi m+r)\omega}\right).\end{aligned} \tag{33}$$

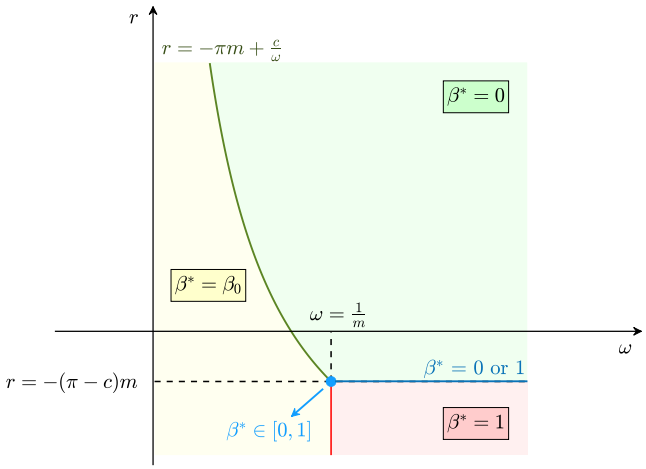


Fig. 4: Regions of worker's optimal adversarial rate $\beta^\star$.

green area marked by $\beta^\star=0$ corresponds to trustworthy computation, while the nonzero-$\beta^\star$ area indicates parameter choices where the worker has an incentive to cheat. The deterrence effect of sampling is not uniform across ω. In the low-sampling regime, i.e., $0<\omega<1/m$, the trust boundary is the curve $r=-\pi m+c/\omega$. Its steep shape near small ω indicates a strong substitution between sampling and escrow. Here, a small increase in ω can sharply reduce the deposit required to deter misbehavior. Fig. 4 also reveals a transition at the point $\omega=1/m$. Once $\omega\geq 1/m$, the deposit threshold no longer decreases. In this case, a larger ω can still raise the detection probability for trustworthy computing, but it also increases the client's verification overhead.

B. Optimal Sampling Strategy

A client can select ω to obtain trustworthy service, i.e., to guarantee that the worker computes with $\beta=0$. Let $\rho=\pi/c$ denote the payment-to-cost ratio. Given $r>-(\pi-c)m$ and $g>\pi$, we establish a sufficient condition under which the client's payoff $u_c(\omega)$ is strictly increasing.

Lemma 1. *Assume that the worker is rational and always follows its best response in Theorem 1. For $0<\omega<c/(\pi m+r)$, the monotonicity of $u_c(\omega)$ can be guaranteed if*

$$G(\omega)\leq c\rho,$$

where $G(\omega)$ is given in (31)*, i.e., $\mathrm{d}u_c(\omega)/\mathrm{d}\omega>0$.*

Proof. By Theorem 1, as the worker uses its best response β_0, the client's payoff becomes

$$\begin{aligned}u_c(\omega)=&gm\left(\frac{c}{(\pi m+r)\omega}\right)^{\frac{1}{m\omega-1}}+r-\omega mc\\ &-(\pi m+r)\left(\frac{c}{(\pi m+r)\omega}\right)^{\frac{m\omega}{m\omega-1}}.\end{aligned} \tag{32}$$

Differentiating this payoff gives (33). In (33), the coefficient of g has the same sign as

$$1-m\omega-m\omega\ln\frac{c}{(\pi m+r)\omega}, \tag{34}$$

on $0<\omega<c/(\pi m+r)$. To determine the sign of (34), we use $\ln t\leq t-1$ to obtain

$$\begin{aligned}&1-m\omega-m\omega\ln\frac{c}{(\pi m+r)\omega}\\ &\geq 1-m\omega-m\omega\left(\frac{c}{(\pi m+r)\omega}-1\right)\\ &=1-\frac{cm}{\pi m+r}>0,\end{aligned} \tag{35}$$

where the last inequality follows from $r>-(\pi-c)m$, so the coefficient of g is positive. Dividing (33) by this coefficient and rearranging shows that $\mathrm{d}u_c(\omega)/\mathrm{d}\omega>0$ is equivalent to $g>G(\omega)$. The relevant $G(\omega)\leq c\rho$ is a sufficient condition for $\mathrm{d}u_c(\omega)/\mathrm{d}\omega>0$ since $g>\pi=c\rho$. □

Lemma 1 gives a sufficient condition for a positive derivative at a given sampling ratio. The following lemma establishes a purely mathematical inequality that we later apply to verify this condition throughout the low-sampling interval and establish monotonicity.

Lemma 2. *For $0<x<y<1$ and $x,y\in\mathbb{R}$, we have*

$$\frac{1-x}{2}+(1-y)\ln\frac{1}{y}-x(1-x)^{2}y^{-\frac{1}{1-x}}>0. \tag{36}$$

Proof. Since $x<y$ and $\ln(1/y)<(1-y)/y$, we have

$$0<\frac{x\ln(1/y)}{1-x}<\frac{x(1-y)}{y(1-x)}<1, \tag{37}$$